PAPER_687: M87* Black Hole Mass Evolution Over Cosmic Time in the UQFF Framework

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Source: `grok_share_fc21e30c24b4.txt`
Domain: SMBH Evolution / Cosmology

Abstract

We simulate the coupled mass evolution of M87* over 13.8 Gyr combining UQFF-modified Bondi accretion $\dot{M}_{\text{Bondi,UQFF}} = \dot{M}_{\text{Bondi}}(\rho_{\text{eff}}/\rho_\infty)(1 + f_{\text{TRZ}})$, suppressed Hawking evaporation, and Blandford-Znajek jet power in UQFF. Using RK4 integration with $\Delta t = 10^{13}$ s steps, we find that M87* grows by $\sim 15\%$ over a Hubble time, consistent with observations suggesting limited current growth.

Primary UQFF Equation

$$\frac{dM}{dt} = \dot{M}_{\text{Bondi,UQFF}} + \dot{M}_{\text{evap,UQFF}} - \frac{P_{\text{jet,UQFF}}}{c^2}$$

Parameters

Parameter	Value	Description
M_0	$6.5 \times 10^9\ M_\odot$	Initial M87* mass
ρ_{ISM}	1.67×10^{-25} kg/m^3	Intracluster medium
T_{ISM}	10^7 K	ICM temperature
T_{Hubble}	13.8 Gyr	Simulation duration

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m^3	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m^3	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day^{-1}	UQFF calibration constant
$[\text{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s^{-1}	Aether oscillation decay

C++ Implementation

```cpp
#include "M87MassEvolutionSimulation.h"

M87MassEvolutionSimulation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import M87MassEvolutionSimulationCalculator

calc = M87MassEvolutionSimulationCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

Walsh et al. 2013 (M87 gas dynamics); Russell et al. 2015 (M87 jet); EHT 2019; UQFF Session 173

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